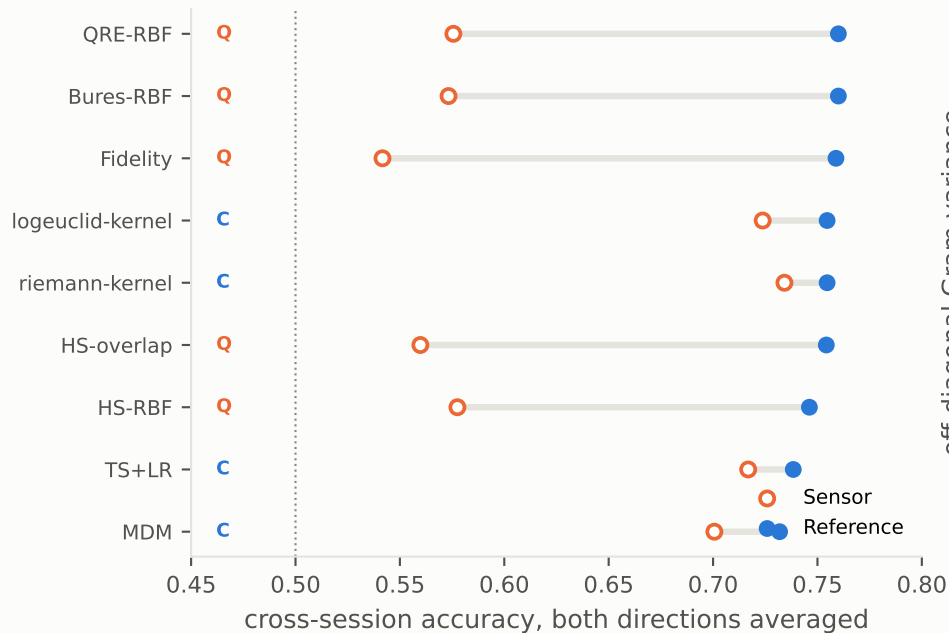
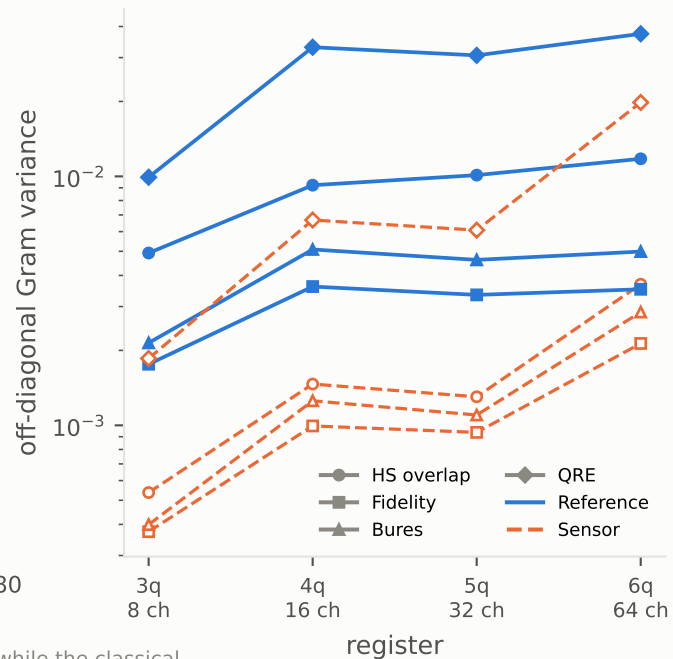


a**b**

(a) Cross-session transfer on IV-2a: in the sensor frame the quantum kernels sit near chance while the classical baselines do not; recentring each session by its own label-free mean closes the gap and leaves every geometry tied.

(b) Gram variance versus register in both frames: in the reference frame no kernel falls below its three-qubit value at any larger register.